Quantum Computing

(23CSC03)

23CSC03	Quantum Computing	3/0/0/3

Nature of the Course: D (Theory Application)

Pre-requisite(s): -

Course Objectives:

1	To introduce the core concepts of classical and quantum information and computation.
2	To familiarize students with the fundamental principles of quantum mechanics applied in quantum computing
3	To explore quantum states, qubit representations, and essential quantum gates.
4	To understand the design and implementation of quantum circuits and operations.
5	To provide introductory knowledge of quantum algorithms and their applications in solving computational problems

Course Outcomes:

Upon completion of the course, students shall have ability to

CO1	Understand the fundamentals of quantum mechanics and represent quantum states	[U]
CO2	Apply quantum postulates to describe and manipulate single and multi-qubit systems.	[AP]
CO3	Construct quantum circuits using quantum gates.	[AP]
CO4	Analyze entanglement and quantum communication protocols, such as superdense coding and teleportation.	[AN]
CO5	Apply basic quantum algorithms, including Deutschs' and Simon's algorithms, to solve computational problems	[AP]
CO6	Apply Quantum Fourier Transform (QFT) in period-finding and explore its role in Shor's factoring algorithm.	[AP]

INTRODUCTION TO QUANTUM MECHANICS	**15**
History of quantum computation and quantum Information- The Circuit Model of Computation – Review of Complex numbers- Vector spaces- -Linear Algebra and Dirac Notation: Dual vectors -Operators-Inner product -outer product- Quantum postulates-Quantum Measurement-Quantum Bits: single Qubits& multiple qubits Diarc and matrix representation- tensor products - Bloch sphere: Representation of a different basis	
QANTUM GATES AND QUANTUM CIRCUITS	**15**
Classical reversible gates and circuits- Single qubit gates and its operation: Pauli gates, Hadamard gate, Rotational gates – superpositions- Multi qubits gates and its operation: CNOT, CCNOT, Swap and Controlled U Gates. Entanglement- Bell state circuits- Quantum Series and Parallel circuits- super dense coding- Quantum Teleportation	
QUANTUM ALGORITHMS	**15**
Introduction to quantum algorithms- Phase kick back- oracle function- Deutsch's Algorithm - Deutsch-Jozsa Algorithm – Simon's Algorithm – Quantum Fourier transform- period findings using QFT- Grover's search algorithm-Shor's factoring algorithm - Case study: Quantum Cryptanalysis of RSA and Symmetric Encryption Using Shor's and Grover's Algorithms	
TOTAL PERIODS	**45**

Quantum Postulates (Axioms)

- Quantum mechanics is built on a set of fundamental postulates.
- These postulates define how quantum systems are described, how they evolve and how measurements are performed.
- Quantum mechanics is built upon a set of fundamental postulates that provide the mathematical framework for describing the physical world at the atomic and subatomic scales.

Old Quantum Theory-Neil Bohr's Postulates

- **Quantum postulates, often referring to Niels Bohr's foundational assumptions in his 1913 atomic model, introduced quantization to explain atomic stability and spectra. These "old quantum theory" ideas paved the way for modern quantum mechanics.**

Neil Bohr's Postulates

Neil Bohr proposed three main postulates for the hydrogen atom.

- **Electrons revolve around the nucleus in fixed, discrete orbits (stationary states) without radiating energy, unlike classical predictions.**
- **Angular momentum in any orbit is quantized:** $L = n\frac{h}{2\pi}$ **, where** n **is a positive integer,** h **is Planck's constant.**
- **Transitions between orbits emit or absorb energy as photons:** $\Delta E = h\nu$**, where** ν **is frequency.**

Modern Quantum Mechanics Postulates

Modern quantum theory has 5-6 core postulates describing states, observables, and measurements.

- **State Postulate:** A system's state is fully described by wavefunction $\psi(\mathrm{r}, t)$; probability density is $|\psi|^2$.
- **Observables Postulate:** Physical quantities correspond to Hermitian operators

(e.g., position $\hat{x} = x$, momentum $\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$).

- **Measurement Postulate:** Outcomes are operator eigenvalues; if ψ is an eigenfunction, measurement yields that eigenvalue.
- **Expectation Value:** Average value is $\langle A \rangle = \int \psi^* \hat{A} \psi \, d\tau$.
- **Time Evolution:** Wavefunction follows time-dependent Schrödinger equation $i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$.
- **Pauli Principle (for fermions):** Total wavefunction is antisymmetric under particle exchange.

Postulate-1: State of a Quantum System

Statement: The state of a quantum system is completely described by a state vector $|\psi\rangle$in a complex Hilbert space $\mathcal{H}$.

Key Points:

- **State vector:** Contains all possible information about the system
- **Hilbert space:** Complete inner product space with complex coefficients
- **Normalization condition:** $\langle\psi | \psi\rangle = 1$
- **Superposition principle:** Linear combinations of states are valid states

Mathematical Details

- **State vector:** $|\psi\rangle = c_1 |1\rangle + c_2 |2\rangle + \ldots + c_n |n\rangle$
- **Probability amplitudes:** $c_i = \langle i | \psi\rangle$
- **Normalization:** $\sum_i |c_i|^2 = 1$

Postulate-2: Observables and Hermitian Operators

Statement:

- **Each measurable physical quantity (observable) corresponds to a Hermitian operator $\hat{A}$ acting on the Hilbert space.**

Key Properties of Observables:

- **Hermitian condition:** $\hat{A} = \hat{A}^\dagger$
- **Real eigenvalues: All possible measurement outcomes**
- **Orthogonal eigenvectors: Form complete basis set**
- **Spectral decomposition:** $\hat{A} = \sum_i a_i \mid \phi_i\rangle\langle\phi_i \mid$

Common Observables:

- **Position:** $\hat{x}$
- **Momentum:** $\hat{p} = -i\hbar \frac{d}{dx}$
- **Energy (Hamiltonian):** $\hat{H}$
- **Angular momentum:** $\hat{L}$

Postulate-3: Measurement and Eigenvalues

Statement:

- **The only possible results of measuring observable $\hat{A}$ are its eigenvalues a_n, where: $\hat{A} \mid \phi_n\rangle = a_n \mid \phi_n\rangle$**

Key Concepts:

- **Eigenvalue equation: Defines allowed measurement values**
- **Discrete spectrum: Finite or countable eigenvalues**
- **Continuous spectrum: Uncountable eigenvalues (like position)**
- **Degenerate eigenvalues: Multiple eigenvectors for same eigenvalue**

Mathematical Framework:

- **Eigenvalue problem: $det(\hat{A} - a_n I) = 0$**
- **Complete set: $\sum_n \mid \phi_n\rangle\langle\phi_n \mid = I$**
- **Orthonormality: $\langle\phi_m \mid \phi_n\rangle = \delta_{mn}$**

Postulate-4: Born Rule (Measurement Probabilities)

Statement:

- The probability of measuring eigenvalue a_n when the system is in state $|\psi\rangle$ is: $P(a_n) = |\langle\phi_n|\psi\rangle|^2$

Expanded Form for Degenerate Cases:

- $P(a_n) = \sum_i |\langle\phi_{n,i}|\psi\rangle|^2$ where $\phi_{n,i}$ are degenerate eigenstates with eigenvalue a_n.

Key Features:

- **Probabilistic nature:** Fundamental randomness in quantum mechanics
- **Born interpretation:** $|c_n|^2$ gives probability
- **Conservation:** $\sum_n P(a_n) = 1$
- **Complex amplitudes:** Phase information crucial for interference

Postulate-5: State Collapse

Statement:

- **After measuring observable $\hat{A}$and obtaining result a_n, the system immediately collapses to the corresponding eigenstate: $|\psi\rangle \rightarrow |\phi_n\rangle$**

Projection Formalism:

- $|\psi_{after}\rangle = \frac{\hat{P}_n|\psi\rangle}{\sqrt{\langle\psi|\hat{P}_n|\psi\rangle}}$
- **Where $\hat{P}_n = |\phi_n\rangle\langle\phi_n|$ is the projection operator.**

Important Notes:

- **Instantaneous process:** **No time evolution during measurement**
- **Irreversible:** **Information about previous superposition lost**
- **Non-unitary:** **Unlike time evolution**
- **Measurement problem:** **Philosophical implications about reality**

Postulate-6: Time Evolution

Statement:

- The time evolution of a quantum system is governed by the time-dependent Schrödinger equation: $i\hbar \frac{\partial}{\partial t} \mid \psi(t)\rangle = \hat{H} \mid \psi(t)\rangle$

Time Evolution Operator:

- For time-independent Hamiltonian: $\mid \psi(t)\rangle = \hat{U}(t) \mid \psi(0)\rangle = e^{-i\hat{H}t/\hbar} \mid \psi(0)\rangle$

Key Properties:

- Unitary evolution: $\hat{U}^\dagger \hat{U} = I$
- Probability conservation: $\langle\psi(t) \mid \psi(t)\rangle = 1$
- Deterministic: Given initial state, future states determined
- Reversible: Time evolution can be undone

Energy Eigenstate Evolution:

- $\mid \psi_n(t)\rangle = e^{-iE_n t/\hbar} \mid \psi_n(0)\rangle$

Real World Applications of Postulates Basis

Field	Application	Postulate Basis
Electronics	Transistors, LEDs	Energy quantization, Pauli exclusion
Optics	Lasers, fiber optics	Stimulated emission from transitions
Medicine	MRI, PET scans	Spin operators, superposition
Computing	Quantum computers	Entanglement, measurement collapse
Navigation	Atomic clocks, GPS	Hyperfine transitions
Energy	Solar panels	Photovoltaic effect, quantized absorption
Microscopy	Electron microscopes	Wave-particle duality
Displays	OLED screens	Bandgap quantization
Materials	Superconducting magnets	Cooper pairs, quantum states
Nanotechnology	Quantum dots (sensors, drugs)	Confinement effects

Limitations and Extensions

Standard Quantum Mechanics Limitations

- Single-particle focus
- Non-relativistic treatment
- Measurement problem unresolved

Extensions

- Quantum Field Theory: Relativistic treatment
- Many-body systems: Entanglement and correlations
- Open systems: Decoherence and dissipation
- Quantum information: Qubits and quantum computation